

Lipschitz-Derivative Triples Beyond Convex Domains

A locally length-space theorem and two sharp obstructions

Candidate partial result for arXiv:2509.20849

13 August 2026

Status: candidate substantial partial proof, likely valid. The packet extends the source's strongest joint necessary condition from locally convex normed domains to locally length metric spaces, proves a quantitative quasiconvex version, and shows both that the source's necessary conditions are not sufficient and that the Baire identity fails on general compact connected metric subspaces. It does not classify all realizable triples. Novelty remains subject to expert literature review.

1 Source question and scope

Maslyuchenko and Wójcicki ask the following inverse problem on source PDF p. 2 [1]:

(see for example [15, 18, 14, 8]). Statements of this type usually involves the set $L(f) = \{x \in \mathbb{R} : \text{Lip } f(x) < \infty\}$. The local Lipschitz derivative $\mathbb{L}ip f$ together with $\text{Lip } f$ characterize the local and pointwise Lipschitzness of functions defined on a metric space [10]. The little Lipschitz derivative was introduced by Cheeger in [6] and together with $\text{Lip } f$ play important role in the research of the first order differential calculus in metric spaces. The crucial fact is the purely metric character of the definitions of Lipschitz derivatives. The above considerations lead to the natural question of characterizing the sets $\ell(f)$, $L(f)$ and $\mathbb{L}(f)$ for functions defined on metric spaces. In recent years, this problem has been investigated in many articles, such as [16, 5, 9]. In particular, in [5] it was shown that $\ell(f)$ is a $G_{\delta\sigma}$ -set and $L(f)$ is an F_{σ} -set for a function $f: \mathbb{R} \rightarrow \mathbb{R}$. In our approach, we introduce generalized notions of semicontinuity of a function and classify the Lipschitz derivatives with help of these properties. This allows us to easily derive the Borel type of sets $\ell(f)$, $L(f)$, $\mathbb{L}(f)$ of a function f acting between arbitrary metric spaces.

Another interesting question to consider is whether for a given triplet of functions (u, v, w) , there exists a continuous function f such that $\text{lip } f = u$, $\text{Lip } f = v$ and $\mathbb{L}ip f = w$. Some advances in this direction were made in [3] and [4]. Our characterization of Lipschitz derivatives in terms of generalized semicontinuity constrains the possible choice of functions (u, v, w) . Moreover, we obtain another necessary criterion for a triple (u, v, w) defined on locally convex subset of a normed space: the upper Baire limits functions of u and v are equal to w .

2. Lipschitz derivatives

Exact transcription. “Another interesting question to consider is whether for a given triplet of functions (u, v, w) , there exists a continuous function f such that $\text{lip } f = u$, $\text{Lip } f = v$ and $\mathbb{L}ip f = w$. Some advances in this direction were made in [3] and [4]. Our characterization of Lipschitz derivatives in terms of generalized semicontinuity constrains the possible choice of functions (u, v, w) . Moreover, we obtain another necessary criterion for a triple (u, v, w) defined on

locally convex subset of a normed space: the upper Baire limits functions of u and v are equal to w .”

For a function g on a metric space, write

$$g^\vee(x) = \inf_{U \ni x} \sup_{y \in U} g(y)$$

for its upper Baire function. The cited necessary criterion is Theorem 7.1 on source PDF p. 16:

7. Baire limit functions of Lipschitz derivatives

For a given function $f: X \rightarrow \overline{\mathbb{R}}$, defined on a metric space X , their *upper Baire function* $f^\vee$ is defined by

$$f^\vee(x) = \inf_{U \in \mathcal{U}(x)} \sup_{u \in U} f(u), \quad x \in X,$$

and their *lower Baire function* $f^\wedge$ is defined by

$$f^\wedge(x) = \sup_{U \in \mathcal{U}(x)} \inf_{u \in U} f(u), \quad x \in X,$$

where $\mathcal{U}(x)$ is the family of all the neighborhoods of x in X . (See, for example, [13].) The upper Baire function $f^\vee$ is upper semicontinuous and the lower Baire function $f^\wedge$ is lower semicontinuous.

A subset D of a normed space X is called *locally convex* if for any point $x \in D$ and their neighborhood U of x in D there is a convex neighborhood V of x in D such that $V \subseteq U$. For example, every convex set and every open set in X is locally convex.

Theorem 7.1. *Let D be a locally convex subset of a normed space X and let $f: D \rightarrow \mathbb{R}$ be a function. Then*

$$(\text{lip } f)^\vee = (\text{Lip } f)^\vee = \mathbb{Lip } f.$$

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We isolate the metric geometry behind this theorem and also test whether the source’s full list of necessary conditions could be sufficient for the inverse problem.

2 Definitions and proof intuition

For a map $f: X \rightarrow Y$ between metric spaces, the three derivatives at x are

$$\begin{aligned} \text{lip } f(x) &= \liminf_{r \downarrow 0} \frac{1}{r} \sup_{y \in B(x, r)} d_Y(f(y), f(x)), \\ \text{Lip } f(x) &= \limsup_{y \rightarrow x} \frac{d_Y(f(y), f(x))}{d_X(y, x)}, \\ \mathbb{Lip } f(x) &= \limsup_{(y, z) \rightarrow (x, x)} \frac{d_Y(f(y), f(z))}{d_X(y, z)}. \end{aligned}$$

The source proves that $\text{lip } f$ is $\mathcal{F}_\sigma$ -lower, $\text{Lip } f$ is $\mathcal{F}_\sigma$ -upper, and $\mathbb{Lip } f$ is upper semicontinuous for continuous f .

Definition 1. Let $C \geq 1$. A metric space X is *asymptotically C -quasiconvex at x* if, for every neighborhood V of x and every $\eta > 0$, there is a neighborhood $U \subseteq V$ of x such that every $y, z \in U$ can be joined by a rectifiable curve contained in V of length at most $(C + \eta)d_X(y, z)$.

Proof Intuition

Convexity in the source proof serves one purpose: it supplies a shortest curve between nearby points. Along an arc-length-parametrized curve c , the little lip of $f \circ c$ is no larger than the little lip of f on the ambient space. The one-dimensional rigidity theorem then integrates this infinitesimal bound along c . Short curves therefore convert the upper Baire envelope of little lip into a bound for the local Lipschitz derivative. Length spaces have curves arbitrarily close to shortest, so the distortion factor tends to one. Two branches that approach tangentially, however, can have very small ambient separation but require a long detour; the local derivative detects the cross-branch quotient while both pointwise derivatives do not.

3 A quantitative metric extension

Lemma 2 (Curve integration). *Let $c : [0, L] \rightarrow X$ be parametrized by arc length and suppose $\text{lip } f(c(t)) \leq \gamma$ for all t . Then*

$$d_Y(f(c(0)), f(c(L))) \leq \gamma L.$$

Proof. Since c is 1-Lipschitz,

$$\sup_{|s-t|<r} \frac{d_Y(f(c(s)), f(c(t)))}{r} \leq \sup_{d_X(y, c(t))<r} \frac{d_Y(f(y), f(c(t)))}{r}.$$

Taking lower limits gives $\text{lip}(f \circ c)(t) \leq \text{lip } f(c(t)) \leq \gamma$. The interval case of Theorem 6.4 in the source gives the conclusion. To see directly that its real-valued interval estimate suffices for an arbitrary metric target, fix $q \in Y$ and apply it to $h_q(t) = d_Y(f(c(t)), q)$; then $\text{lip } h_q \leq \text{lip}(f \circ c)$. Taking $q = f(c(0))$ gives the displayed inequality. $\square$

Theorem 3 (Quantitative Baire-envelope bound). *Suppose X is asymptotically C -quasiconvex at x . For every continuous map $f : X \rightarrow Y$ into any metric space,*

$$(\text{lip } f)^\vee(x) \leq (\text{Lip } f)^\vee(x) \leq \mathbb{L} \text{ip } f(x) \leq C(\text{lip } f)^\vee(x). \quad (1)$$

Proof. The first two inequalities follow from $\text{lip } f \leq \text{Lip } f \leq \mathbb{L} \text{ip } f$ and upper semicontinuity of $\mathbb{L} \text{ip } f$. Put $A = (\text{lip } f)^\vee(x)$. If $A = \infty$, there is nothing to prove. Choose $\gamma > A$. Some neighborhood V of x satisfies $\text{lip } f < \gamma$ throughout V . Given $\eta > 0$, choose $U \subseteq V$ from asymptotic quasiconvexity. Lemma 2 applied to the joining curves gives

$$d_Y(f(y), f(z)) \leq \gamma(C + \eta)d_X(y, z), \quad y, z \in U.$$

Thus $\mathbb{L} \text{ip } f(x) \leq \gamma(C + \eta)$. Let first $\gamma \downarrow A$ and then $\eta \downarrow 0$. $\square$

Corollary 4 (Locally length domains). *If X is locally a length space, then for every metric space Y and every continuous $f : X \rightarrow Y$,*

$$(\text{lip } f)^\vee = (\text{Lip } f)^\vee = \mathbb{L} \text{ip } f.$$

Proof. Fix x , a neighborhood V , and $\eta > 0$. Work in a length-space neighborhood of x and choose a small ball $B(x, R) \subseteq V$. If $y, z \in B(x, R/4)$, a curve of length at most $(1 + \min\{\eta, 1/4\})d_X(y, z)$ remains in $B(x, R)$: its distance from x is bounded by the distance from x to y plus the length already traversed. Hence X is asymptotically 1-quasiconvex at x , and Theorem 3 applies. $\square$

This includes geodesic spaces, Riemannian and Finsler manifolds, metric graphs, CAT spaces, and open subsets of length spaces. It also allows arbitrary metric codomains, whereas source Theorem 7.1 is stated for real-valued maps.

4 Two obstructions for the inverse problem

Proposition 5 (The source conditions are not sufficient). *On $I = [-1, 1]$, let*

$$u = v = w = \mathbf{1}_{\{0\}}.$$

This triple satisfies $u \leq v \leq w$, the source's respective $\mathcal{F}_\sigma$ -lower, $\mathcal{F}_\sigma$ -upper, and upper semicontinuity conditions, and $u^\vee = v^\vee = w$. Nevertheless, no continuous $f : I \rightarrow \mathbb{R}$ realizes $(u, v, w) = (\text{lip } f, \text{Lip } f, \mathbb{Lip } f)$.

Proof. The singleton is closed, its complement is open (hence F_σ in a metric space), and w is upper semicontinuous; these observations verify all the stated regularity and envelope conditions.

If such f existed, then $\text{lip } f \leq 1$. The interval case of source Theorem 6.4 would make f 1-Lipschitz, hence absolutely continuous and differentiable almost everywhere. At every differentiability point $x \neq 0$, source Theorem 3.2 gives $|f'(x)| = \text{Lip } f(x) = 0$. Thus $f' = 0$ almost everywhere, so f is constant. Its three Lipschitz derivatives are then zero at the origin, a contradiction. $\square$

Theorem 6 (Tangential double-arc counterexample). *For every $\alpha > 1$, set*

$$X_\alpha = \{(t, t^\alpha), (t, -t^\alpha) : 0 \leq t \leq 1\} \subset \mathbb{R}^2$$

with the Euclidean subspace metric, let $o = (0, 0)$, and let $f(x, y) = y$. Then X_α is compact and connected, while

$$(\text{lip } f)^\vee(o) = (\text{Lip } f)^\vee(o) = 0, \quad \mathbb{Lip } f(o) = 1. \quad (2)$$

Proof. The coordinate projection f is globally 1-Lipschitz. At the origin, the pointwise quotient along either branch is

$$\frac{t^\alpha}{\sqrt{t^2 + t^{2\alpha}}} \rightarrow 0,$$

so $\text{lip } f(o) = \text{Lip } f(o) = 0$. At a nonzero point of either smooth branch, all pointwise derivatives equal the norm of the vertical coordinate on the unit tangent,

$$m_\alpha(t) = \frac{\alpha t^{\alpha-1}}{\sqrt{1 + \alpha^2 t^{2\alpha-2}}} \rightarrow 0 \quad (t \downarrow 0).$$

This proves both upper-envelope equalities in (2).

For the opposite points $y_t = (t, t^\alpha)$ and $z_t = (t, -t^\alpha)$,

$$\frac{|f(y_t) - f(z_t)|}{\|y_t - z_t\|_2} = \frac{2t^\alpha}{2t^\alpha} = 1.$$

They converge to o , so $\mathbb{Lip } f(o) \geq 1$; global 1-Lipschitzness gives the reverse inequality. Thus (2) holds. $\square$

The failure is exactly a path-distortion phenomenon. Every path in X_α joining y_t to z_t passes through o , so its length is at least $2t$, whereas the ambient distance is $2t^\alpha$. Their ratio $t^{1-\alpha}$ diverges.

5 Verification, limitations, and novelty

The accompanying verifier evaluates the radial quotient, the exact tangent slope, the cross-branch quotient, and the path/chord distortion for $\alpha = 3/2, 2, 3$ down to $t = 10^{-9}$. It independently compares the tangent formula with centered finite differences. Every check passed; the largest finite-difference error was below $2.6 \cdot 10^{-11}$.

Scope limitation. Proposition 5 rules out the obvious sufficiency conjecture, and Corollary 4 resolves the domain-geometry component of the strongest joint constraint. The full inverse

problem remains open: even on an interval, realizability includes additional measure-theoretic and integrability constraints not captured by this argument.

Bounded novelty audit. The run indexes and primary-source arXiv searches through 13 August 2026 were checked using arXiv:2509.20849 and the phrases “little Lipschitz derivative,” “upper Baire,” “length space,” and “lower scaled oscillation.” Searches found the source, arXiv:1910.14527, and recent Takagi–van der Waerden constructions, but no locally length extension, quantitative quasiconvex bound, singleton-spike obstruction, or tangential-double-arc example. Novelty is plausible, not certified.

Human-review recommendation: send to a metric-analysis expert. The highest-value checks are the metric-codomain scalarization in Lemma 2, the neighborhood bookkeeping in Corollary 4, and the assertion that the singleton indicator meets the source’s generalized semicontinuity definitions.

References

- [1] O. V. Maslyuchenko and Z. M. Wójcicki, *Classification of Lipschitz derivatives in terms of semicontinuity and the Baire limit functions*, arXiv:2509.20849 (2025).